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Program

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#include<stdio.h>
#include<conio.h>
#include<malloc.h>
struct node
{
        int data;
        struct node *next;
};
struct node *start=NULL;
struct node *create(struct node *start);
struct node *display(struct node *start);
struct node *insert_beg(struct node *start);
struct node *insert_end(struct node *start);
struct node *del_beg(struct node *start);
struct node *del_end(struct node *start);
struct node *count(struct node *start);
struct node *search(struct node *start);
int main()
{
        int op;
        do
        {
                printf("\n 1.Create link list");
                printf("\n 2.Display link list");
                printf("\n 3.Insert at beggining oflink list");
                printf("\n 4.Insert at end of link list");
                printf("\n 5.Delete at beggining of link list");
                printf("\n 6.Delete at end of link list");
                printf("\n 7.Count total nodes");
                printf("\n 8.Search a node");
                printf("\nEnter your option ");
                scanf("%d",&op);
                switch (op)
                {
                        case 1:
                                start=create(start);
                                printf("\n Link List is created");
                                break;
                        case 2:
                           display(start);
                           break;
                        case 3:
                           start=insert_beg(start);
                           printf("\n Node inserted at beggining");
                           break;
                        case 4:
                           start=insert_end(start);
                           printf("\n Node inserted at end");
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                        break;
                    case 5:
                        start=del_beg(start);
                        printf("\n Node deleted at beggining");
                        break;
                    case 6:
                        start=del_end(start);
                        printf("\n Node deleted at end");
                        break;
                    case 7:
                        count(start);
                        break;
                    case 8:
                        search(start);
                        break;
                default:
                    printf("\n Invalid option");
                    break;
                }
        }
        while(op!= 16);
        return 0;
}
struct node *create(struct node *start)
{
        struct node *new_node ,*ptr ;
        int num;
        printf("\n Enter -1 to stop the link list creation");
        printf("\n Enter link list data ");
        scanf("%d", &num);
        while(num != -1)
        {
         new_node = (struct node *)malloc(sizeof(struct node));

    if (new_node == NULL)
    {
        printf("\nMemory allocation failed");
        return start;
    }

    new_node->data = num;
    new_node->next = NULL;

    if (start == NULL)
    {
        start = new_node;
    }
    else
    {
        ptr = start;
```

```
            while (ptr->next != NULL)
            {
                ptr = ptr->next;
            }

            ptr->next = new_node;
        }
                    printf("Enter linked list data: ");
        scanf("%d", &num);
            }
            return start;
}
struct node *display(struct node *start)
{
            struct node *ptr;
            ptr = start;
             if (ptr == NULL)
    {
        printf("\nLinked list is empty");
        return start;
    }
            while(ptr != NULL)
            {
                    printf("\n %d",ptr->data);
                    printf("\t %d",(void *)ptr);
                    printf("\t %d",(void *)ptr->next);
                    ptr=ptr->next;
            }
            return start;
}
struct node *insert_beg(struct node *start)
{
    struct node *new_node;
    int num;
    printf("\nEnter data to insert: ");
    scanf("%d", &num);
    new_node = (struct node *)malloc(sizeof(struct node));
    new_node->data = num;
    new_node->next = start;
    start = new_node;
    return start;
}
struct node *insert_end(struct node *start)
{
    struct node *new_node ,*ptr ;
            int num;
            printf("\n Enter link list data");
            scanf("%d", &num);
            new_node = (struct node *)malloc(sizeof(struct node));
            new_node->data = num;
             ptr = start;
```

```
        while (ptr->next != NULL)
        {
            ptr = ptr->next;
        }

        ptr->next = new_node;
         new_node->next = NULL;
         return start;
}
struct node *del_beg(struct node *start)
{   struct node*ptr;
        if (start==NULL)
        {
                printf("\n UNDERFLOW");
        }
        ptr = start;
        start=start->next;
        free(ptr);
        return start;
}
struct node *del_end(struct node *start)
{
        struct node *ptr, *preptr;
        if (start==NULL)
        {
                printf("\n UNDERFLOW");
        }
        ptr = start;
        while(ptr->next != NULL)
        {
                preptr = ptr;
                ptr= ptr->next;
        }
        preptr->next=NULL;
        free(ptr);
        return start;

}
struct node *count(struct node *start)
{
        struct node *ptr;
        int c=0;
        if (start==NULL)
        {
                printf("\n UNDERFLOW");
        }
        else
        {
                ptr = start;
```

```
        while (ptr->next != NULL)
        {
         c++;
        ptr = ptr->next;
        }
      printf("\n Total number of nodes are %d",c);
        }
        return start;
}
struct node *search(struct node *start)
{
        struct node *ptr;
        int k;
        ptr=start;
        if (start==NULL)
        {
                printf("\n UNDERFLOW");
        }
        else
        {
                printf ("\n Enter the number you want to search : ");
                scanf("%d", &k);
                ptr = start;

        while (ptr->data != k)
        {
           ptr = ptr->next;
        }
                if (ptr->data == k)
                {
                printf("\n Number found: %d", ptr->data);
                printf("\t Address of node: %d",(void *)ptr);
                printf("\t Address of next node: %d", (void *)ptr->next);
                }
                else
                {
                        printf("\n Number not found");
                }
        }
        return start;
}
```

Output

1.Create link list
2.Display link list
3.Insert at beggining oflink list
4.Insert at end of link list
5.Delete at beggining of link list
6.Delete at end of link list
7.Count total nodes
8.Search a node

```
Enter your option 1

 Enter -1 to stop the link list creation
 Enter link list data 12
Enter linked list data: 21
Enter linked list data: 22
Enter linked list data: 23
Enter linked list data: 24
Enter linked list data: -1

 Link List is created
 1.Create link list
 2.Display link list
 3.Insert at beggining oflink list
 4.Insert at end of link list
 5.Delete at beggining of link list
 6.Delete at end of link list
 7.Count total nodes
 8.Search a node
Enter your option 2

 12      12479248        12479280
 21      12479280        12479312
 22      12479312        12479344
 23      12479344        12479376
 24      12479376        0
 1.Create link list
 2.Display link list
 3.Insert at beggining oflink list
 4.Insert at end of link list
 5.Delete at beggining of link list
 6.Delete at end of link list
 7.Count total nodes
 8.Search a node
Enter your option 3

Enter data to insert: 30

 Node inserted at beggining
 1.Create link list
 2.Display link list
 3.Insert at beggining oflink list
 4.Insert at end of link list
 5.Delete at beggining of link list
 6.Delete at end of link list
 7.Count total nodes
 8.Search a node
Enter your option 4

 Enter link list data20
```

Node inserted at end
1.Create link list
2.Display link list
3.Insert at beggining oflink list
4.Insert at end of link list
5.Delete at beggining of link list
6.Delete at end of link list
7.Count total nodes
8.Search a node
Enter your option 5

Node deleted at beggining
1.Create link list
2.Display link list
3.Insert at beggining oflink list
4.Insert at end of link list
5.Delete at beggining of link list
6.Delete at end of link list
7.Count total nodes
8.Search a node
Enter your option 6

Node deleted at end
1.Create link list
2.Display link list
3.Insert at beggining oflink list
4.Insert at end of link list
5.Delete at beggining of link list
6.Delete at end of link list
7.Count total nodes
8.Search a node
Enter your option 7

Total number of nodes are 5
1.Create link list
2.Display link list
3.Insert at beggining oflink list
4.Insert at end of link list
5.Delete at beggining of link list
6.Delete at end of link list
7.Count total nodes
8.Search a node
Enter your option 8
Enter the number you want to search : 23
Number found: 23 Address of node: 12479344 Address of next node: 12479376